Realize the full potential of AI with SAP Databricks in Business Data Cloud



databricks

SAP

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Innovation starts with a trusted data foundation

AI can lead to transformative opportunities for modern enterprises, but many organizations are quickly discovering that they lack the data foundation needed to support and scale AI solutions. Poor data quality remains a major challenge, with AI developers often spending more time cleaning and preparing data than building models and generating insights.

One of the key reasons organizations struggle with data quality is the continued reliance on traditional practices—like extracting data from SAP applications and rebuilding it elsewhere—leading to fragmented context, duplicated work, quality issues, and ultimately a lack of trust in data and AI. Today, data is no longer just a byproduct of business applications; it's a strategic asset that drives innovation and influences every decision. Yet this persistent lack of trust between businesses and the data they rely on remains a barrier to true innovation in data and AI-driven corporate transformation.



Breakthroughs in AI begin with connected data and applications

To succeed with AI, organizations need an end-to-end strategy that connects applications, data, and AI—a strategy that SAP is uniquely positioned to support. This approach starts with SAP applications that run critical business processes and generate the data needed to fuel AI and innovation.

This data is centralized and enriched in SAP Business Data Cloud—a fully managed software-as-a-service (SaaS) solution that unifies and governs all SAP data and connects it with third-party data to give line-of-business leaders context to make more impactful decisions. Because Business Data Cloud is deeply integrated across SAP applications, critical data retains its original business context and semantics—which enables AI to deliver reliable, responsible, and relevant results.

Business Data Cloud, as a complete SaaS solution for data and analytics, delivers fully managed capabilities for creating a business data fabric that supports business warehouse modernization, planning, data engineering, and AI/ML workloads. Most importantly, the natural integration provided by data products between Business Data Cloud and your SAP applications means that data retains its original business context and semantics. Along with these data products, SAP also delivers intelligent applications that help uncover hidden insights and fast-track decision-making out of the box.

SAP and Databricks: Redefining how applications and data platforms work together

Through a landmark partnership with Databricks, SAP has integrated the power of Databricks Data Intelligence Platform directly into Business Data Cloud. SAP Databricks is a fully SAP-managed solution that combines industry leading AI/ML, data science, and data engineering capabilities with semantically rich, business-ready data from SAP applications. This integrated solution empowers data and AI users to create advanced AI solutions, execute real-time SQL analytics, and run data processing workloads on rich SAP data in a pro-code development environment.

Key SAP Databricks capabilities:

- **Unify data with zero copies:** Use Delta Sharing to effortlessly connect to contextual SAP data products and blend with third-party data—without managing pipelines and copying data.
- **Deliver trusted data products:** Tap into the data product economy to fuel data, analytics, and AI initiatives with curated data products.
- **Advanced SQL analytics:** Analyze data at scale to unlock real-time, on-the-fly insights with serverless Databricks SQL.
- **Develop with pro-code tooling:** Use Databricks Notebook to write Spark pipelines, run SQL analysis, and create ML models.
- **Build custom AI and ML:** Connect curated, AI-ready data with a powerful suite of data science capabilities.

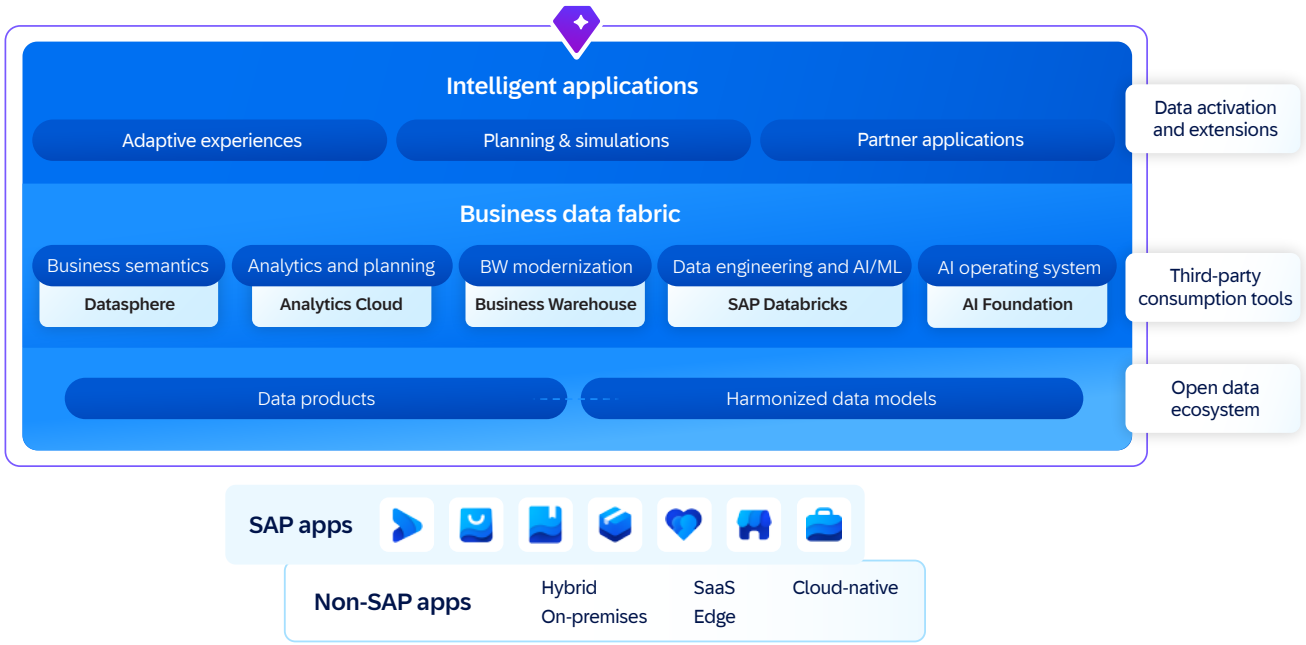


Figure 1: Architecture of SAP Business Data Cloud



AI is transforming every area of business—from finance to human resources to supply chains.

Explore these three use cases to see how SAP Databricks helps leaders across departments put trusted business data to work and drive real impact across their organizations.

Use case for chief financial officers: Financial planning and forecasting

Chief financial officers (CFOs) face a broad range of challenges in today's volatile and uncertain market. Traditional tools like manual spreadsheets and static quarterly reports lack the ability to reflect real-time changes, making them ineffective in today's fast-moving business environment. Unpredictable customer demand, fluctuating supplier costs, and inconsistent payment cycles further complicate efforts. To steer their organizations effectively, CFOs need accurate projections and reliable financial forecasts that enable them to plan ahead with confidence. Just as important is the ability to stay agile and respond quickly and decisively as conditions change. That's why many CFOs are turning to smarter, more flexible planning tools—ones that pull in real-time data, highlight trends as they emerge, and help anticipate challenges before they become problems that impact their businesses.



Solution

Business Data Cloud delivers finance data products from SAP Cloud ERP and provides ready-to-use planning templates and simulation capabilities. With SAP Databricks, you can combine this SAP finance data with external sources to build advanced financial predictive models.

- **Intelligent applications:** The Finance Intelligence package in Business Data Cloud offers an out-of-the-box solution for examining receivables, payables, and inventory to uncover hidden cash opportunities.
- **Scenario modeling and simulation:** Finance teams can simulate the impact of different business decisions directly within SAP Analytics Cloud.
- **SAP managed data products:** Deep integration with SAP ERP enables zero-copy access to accounts payable and accounts receivable, as well as general ledger data.
- **Joule copilot:** Joule enables natural-language questions like “Will I have enough cash flow to support our new growth initiative?”
- **SAP Databricks:** Finance teams can develop predictive models to forecast cash flow and share them as derived data products within Business Data Cloud.

Business outcome

This scenario demonstrates how Business Data Cloud, combined with SAP Databricks, enables the development of AI-powered solutions that empower CFOs to make informed decisions, drive growth initiatives, invest in innovation, and lead business transformation.

With SAP solutions, organization can achieve:

- **Confident and strategic decision-making:** Confident cash flow projections enable leadership to make informed investments in growth, innovation, and expansion with reduced financial risk.
- **Improved financial stability to support business growth:** Accurate forecasting minimizes surprises, reduces the need for short-term borrowing, and ensures consistent liquidity to meet obligations.
- **Greater agility and resilience:** With clear visibility into cash positions, companies can quickly adapt to market changes, seize new opportunities, and navigate economic uncertainty more effectively.

Equipped with these dynamic insights, CFOs can confidently present a range of potential outcomes—highlighting risks, uncovering opportunities, and showcasing data-driven agility to their executive teams.

Use case for chief human resource officers: Workforce composition analysis

Chief human resource officers (CHROs) and HR leaders from all levels are starting to play more strategic roles within their organizations. As such, they face various critical internal and external challenges—like skills shortages, high external hiring costs, and evolving employee expectations. Meeting these challenges requires HR leaders to adapt to a new era of AI. For example, the introduction of AI into HR is driving significant transformation and reshaping how HR professionals manage talent, streamline processes, and make data-driven decisions for recruitment and talent acquisition. With the volatility of the labor market, HR leaders need to make sure to maintain high employee engagement rates. In addition, HR also must ensure productivity, workplace expectations, and the well-being of employees.



Solution

Business Data Cloud can seamlessly integrate with SAP SuccessFactors for data such as employee profiles, role descriptions, skills, and performance metrics, along with insights from open job requisitions. Leveraging SAP Databricks, these data products can be combined with external HR insights and other third-party data.

- **Intelligent applications:** The People Intelligence package in Business Data Cloud gives real-time visibility into the total workforce and enables actionable insights that remove the guesswork from workforce planning.
- **Data Products:** Real-time integration with SAP SuccessFactors and Business Data Cloud enables zero-copy access to SAP Data Products within the HR domain.
- **Joule copilot:** Joule supports recruiting and employee growth by enabling the development of learning programs and hiring strategies.
- **SAP Databricks:** SAP Databricks blends internal workforce data with external sources using Delta Sharing and collaborative notebooks.

Business outcome

This scenario showcases how Business Data Cloud with SAP Databricks and SAP Business AI empowers CHROs to make informed, strategic decisions that drive workforce resilience and organizational growth.

With SAP solutions, organizations can achieve:

- **Optimized human resource management:** A holistic view of the workforce enables data-driven, forward-looking decisions about hiring, upskilling, and talent allocation and ensure workforce planning aligns with business goals.
- **Improved human resource operations:** Skill gaps can be identified and targeted hiring and learning strategies can be enacted to enhance recruitment competitiveness and speed up time-to-hire for in-demand roles.
- **Increased workforce engagement:** AI agents personalize the employee experience using a 360-degree model.

Using insights from AI, HR leaders can take a data-driven approach to workforce planning by strategically enhancing the effectiveness of their most important asset—their people.

Use case for chief operating officers: Supply chain risk management

Chief operating officers (COOs) are focused on fueling growth by building the agility and resilience needed to navigate unprecedented challenges. To do this, supply chain leaders must digitally transform business processes and help employees work more effectively with AI. However critical challenges—both internal and external—stand in the way of progress. Externally, supply chain disruptions, global conflicts, regulations, and the race to carbon zero pose significant risks. Internally, data silos and poor data quality—often due to multiple legacy systems and manual, error-prone processes—cause significant challenges.

To stay ahead of disruption, COOs must maximize the vast amount of data produced by their organizations to enable smarter decisions and transformative outcomes.



Solution

Business Data Cloud provides advanced analytics and AI-driven insights specifically tailored to supply chain operations by connecting to SAP applications and delivering curated, supply-chain-relevant data products. These data products can be shared to SAP Databricks for use in building and deploying custom AI.

- **Intelligent applications:** The Supply Chain Intelligence package in Business Data Cloud facilitates collaboration between supply chain and procurement to diversify suppliers, adjust product schedules, and increase safety stock of critical components.
- **Data Products:** Supply chain data products combine real-time data from SAP and third-party sources to provide a unified, contextualized view of operations.
- **Business AI:** Business AI tools generate comprehensive vendor risk profiles to inform sourcing decisions and strengthen resilience.
- **SAP Databricks:** The SAP Databricks pro-code environment enables the development of predictive models that feed operational applications and analytics with advanced insights for real-time supply chain optimization.

Business outcome

This scenario illustrates how COOs can use Business Data Cloud to take proactive steps towards shifting from reactive responses to risk mitigation for a more resilient and agile supply chain.

With SAP solutions, organizations can achieve:

- **Risk identification:** Better decisions can be made regarding the use of partners within the supply chain by identifying potential risks in advance.
- **Situational awareness:** Supply chain data products incorporate transactional data along with external data—such as weather patterns or geopolitical risks—for a more complete understanding of the factors affecting continuity.
- **Operational elasticity:** Historical disruption patterns, current supply chain signals, and external risk factors can be evaluated to identify alternate suppliers from SAP ISBN, estimate potential production delays, and quantify the financial impact of disruptions.

With a trusted data foundation fueling AI, COOs can take proactive steps to diversify suppliers, adjusting production schedules, and adjust the safety stock of critical materials.

Building AI you can trust

Realize the full potential of AI with SAP Databricks in Business Data Cloud

Native capabilities within Business Data Cloud enable it to access all application data and semantics across an enterprise. For SAP applications, prebuilt data products with SAP-delivered content further accelerate time to value. Business Data Cloud also natively connects to any source—including on-premises SAP applications, third-party applications, data warehouses, and data lakes.

Once the data is available, no additional duplication is needed. Data products containing full business context are made available with Delta Sharing. Data scientists and AI engineers can use the pro-code environment to produce AI/ML models for inclusion back into the business data fabric. Intelligent applications, analytics, agentic AI, or any other use case can consume the results to enable data driven decision making.

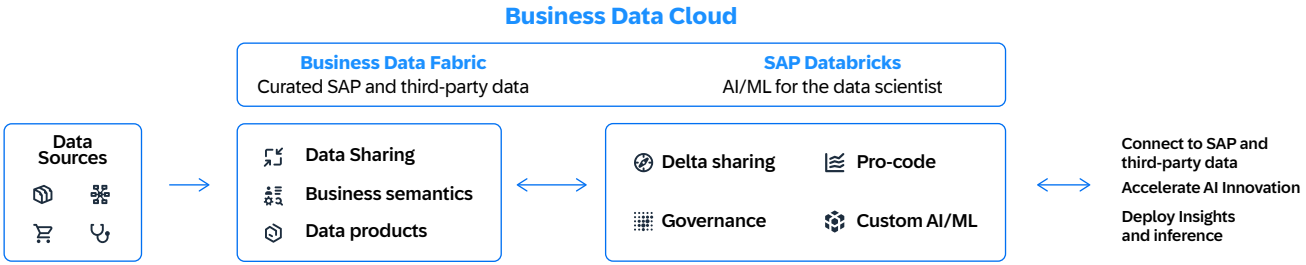


Figure 2: Integration between Business Data Cloud and SAP Databricks

How to deploy your own AI use case in SAP Databricks

Business Data Cloud unites your entire data team to transform raw data into intelligent business solutions. This process starts with data engineers seamlessly connecting SAP and non-SAP sources—without duplicating data. Then a unified, consistent semantic layer is created to bring clarity across systems.

With this trusted foundation in place, data scientists can build, train, and deploy AI and ML models that deliver the insights needed for smarter decision-making—by both human users and AI agents. These insights are then put into action by business and data analysts who create solutions that power real digital transformation.



This flexible framework can be applied to virtually any business scenario. Here's a step-by-step look at how to launch your own AI use case:

1. In Business Data Cloud, subscribe to the relevant intelligent application for access to SAP-managed data products.
2. Install the data package containing data products.
3. Initiate Delta Sharing of appropriate data products to SAP Databricks.
4. Integrate and model additional data necessary for the use case. Provision SAP Databricks as Delta Sharing recipient.
5. In SAP Databricks, develop a Python notebook to blend and transform the incoming data. Manage the transformed dataset using the Feature Store.
6. Experiment, train, and validate the model using algorithms and manage the trained model using Unity Catalog.
7. Deploy the trained model as a Model Serving Endpoint for real-time application consumption.
8. Schedule batch inference.
9. Generate profile and drift tables and build SAP Databricks alerts on top of the tables for automated notification and actions.
10. Consume the model endpoints in a custom application using protocols such as REST.
11. For additional analytic and planning use cases:
    - Delta Share the inference result to Business Data Cloud for Datasphere and Analytics Cloud consumption.
    - In Datasphere, create an Analytical Model with the data products from Databricks.
    - Build an Analytics Cloud dashboard with the Analytical Model to visualize the result.

Meeting the needs of all AI and data professionals

Unleash the full value of your business data with SAP Databricks in Business Data Cloud. SAP Databricks unifies business-ready data with market-leading AI/ML, data science, and data engineering to enable unparalleled business insights and AI potential.

Business Data Cloud

- **Intelligent applications:** Automate, adapt, and learn in real time with AI-powered applications that understand your business.
- **SAP-managed data products:** Use fully managed, curated data products across all lines of business and eliminate the hidden costs of data extraction to save time, resources, and effort.
- **Data harmonization:** Harmonize all your mission-critical data with an open data ecosystem and gain unmatched knowledge of your business with a powerful semantic layer.

SAP Databricks

- **Domain-specific generative AI solutions:** Develop cost-effective generative AI solutions tailored to your unique business needs and deliver more accurate results for informed decisions.
- **Data-driven decision-making:** Build AI/ML solutions on AI-ready SAP data blended with third-party data to enhance decision making.
- **Large-scale ad-hoc data analysis:** Analyze and explore huge amounts of data by using on-demand SQL to uncover dynamic insights that help drive and support agile decision-making.

Additional resources



Explore SAP Databricks



Explore Business Data Cloud



Try Business Data Cloud now

SAP Bring out your best.

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